

VERIXA Platform — Complete UML System Specifications & Architectural Diagrams

Project: VERIXA AI-Safe Social Media Platform & Neural Defense Architecture
Repository: khabir-278/verixa
Specification Standard: Unified Modeling Language (OMG UML 2.5)
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1. Executive Architectural Overview

VERIXA is an end-to-end, AI-defended social platform engineered around **zero-tolerance digital safety, transparent feed recommendation, and multi-signal behavioral risk management.**

The system architecture is bifurcated into:

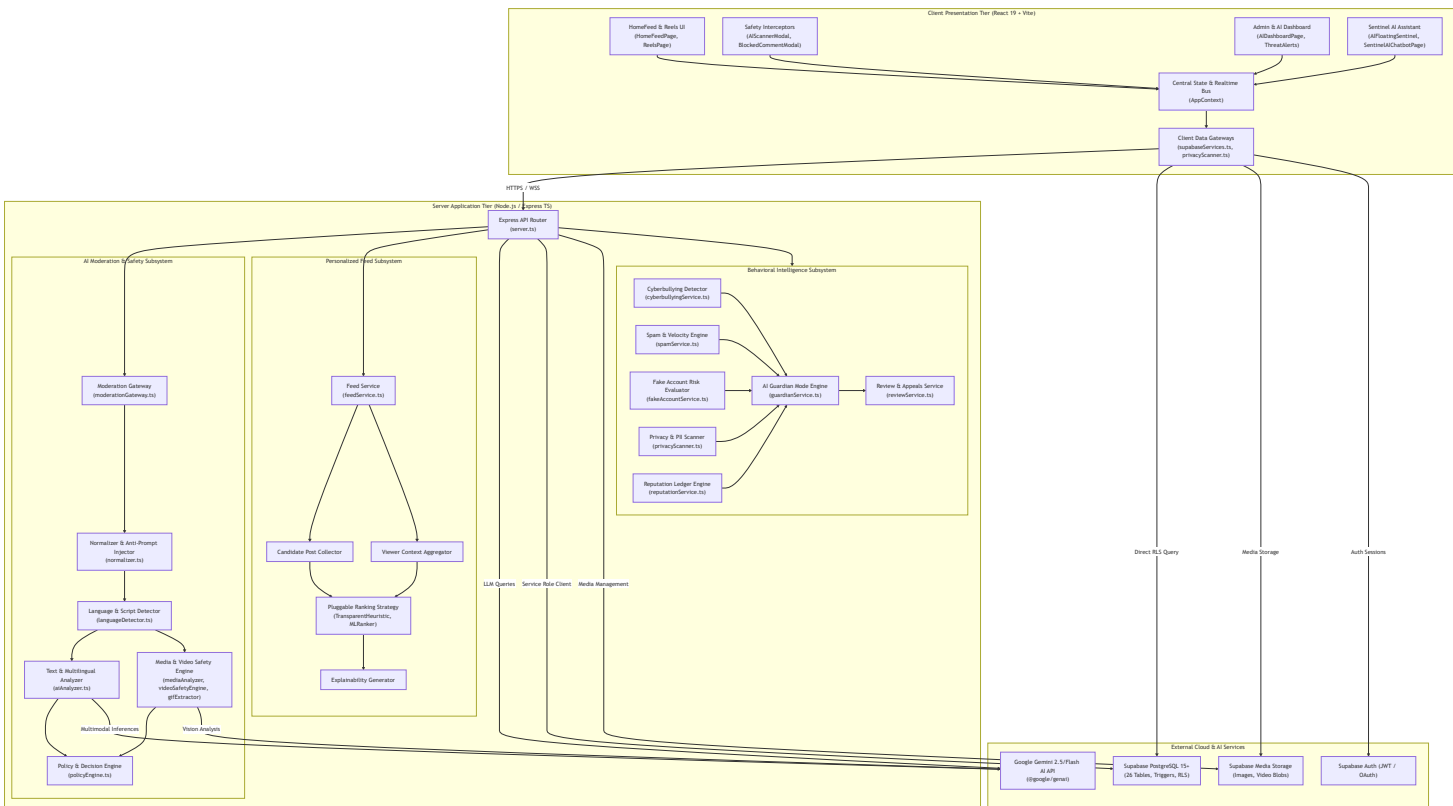
- **Client Tier:** A responsive single-page application built on **React 19, TypeScript, Tailwind CSS 4,** and **Lucide/Motion** components, communicating with backend micro-services via REST and Supabase Realtime subscriptions.

- **Server Tier:** An Express.js and TypeScript runtime exposing specialized engines:
 - i. **Centralized AI Moderation Gateway** (ModerationGateway)
 - ii. **Multilingual & Script Analyzer** (languageDetector , aiAnalyzer)
 - iii. **Media Safety & Deepfake Detector** (mediaAnalyzer , videoSafetyEngine , gifExtractor)
 - iv. **Behavioral Safety Suite** (Cyberbullying, Spam, Fake Account Heuristics, Privacy Scanner)
 - v. **AI Guardian Mode Engine** (guardianService)
 - vi. **Human Review & Appeals System** (reviewService)
 - vii. **Personalized Feed Recommendation Engine** (feedService , TransparentHeuristicRanker , MLRecommendationRanker)
 - viii. **Sentinel AI Chatbot Assistant** (GoogleGenAI integration)
- **Data Tier:** Supabase (PostgreSQL 15+) with Row-Level Security (RLS), real-time replication, storage buckets, database triggers, and synchronized fallback local JSON event ledgers.

2. Structural UML Diagrams

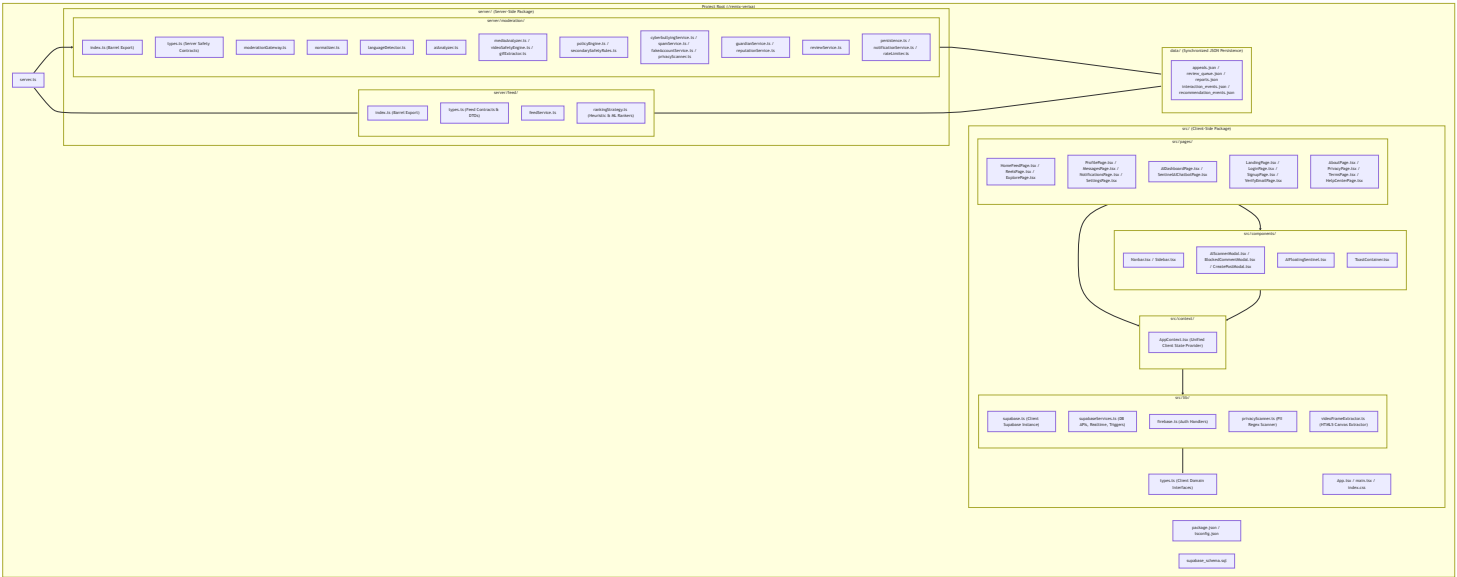
2.1 High-Level Component Diagram

This diagram captures the physical and logical boundaries of VERIXA's subsystem components, inter-component interfaces, and external API dependencies.



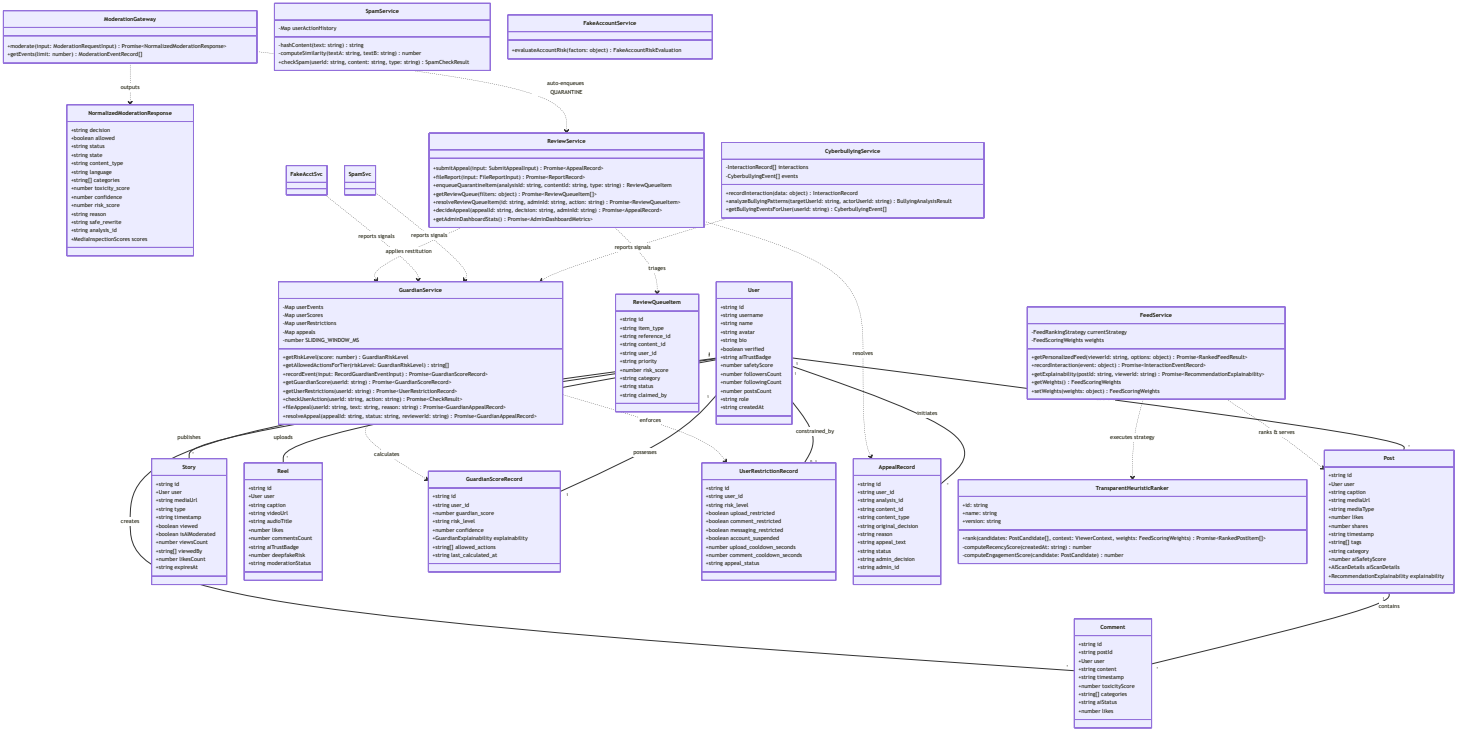
2.2 Modular Package Diagram

This diagram displays the structural organization of source code packages, module boundaries, and dependency hierarchies.



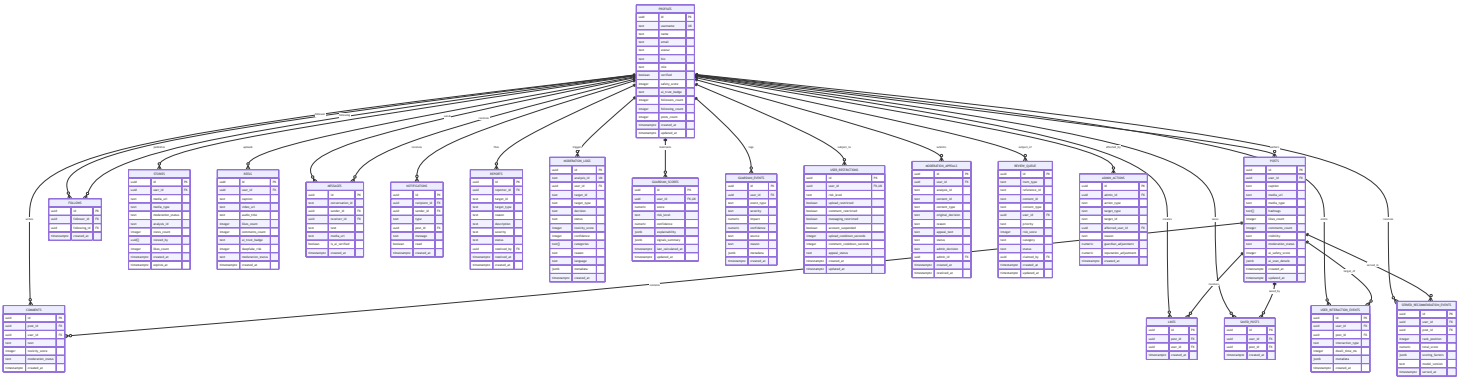
2.3 Detailed Domain & Service Class Diagram

This class diagram depicts the domain entities, safety records, controllers, and core services with their properties, method signatures, and associations.



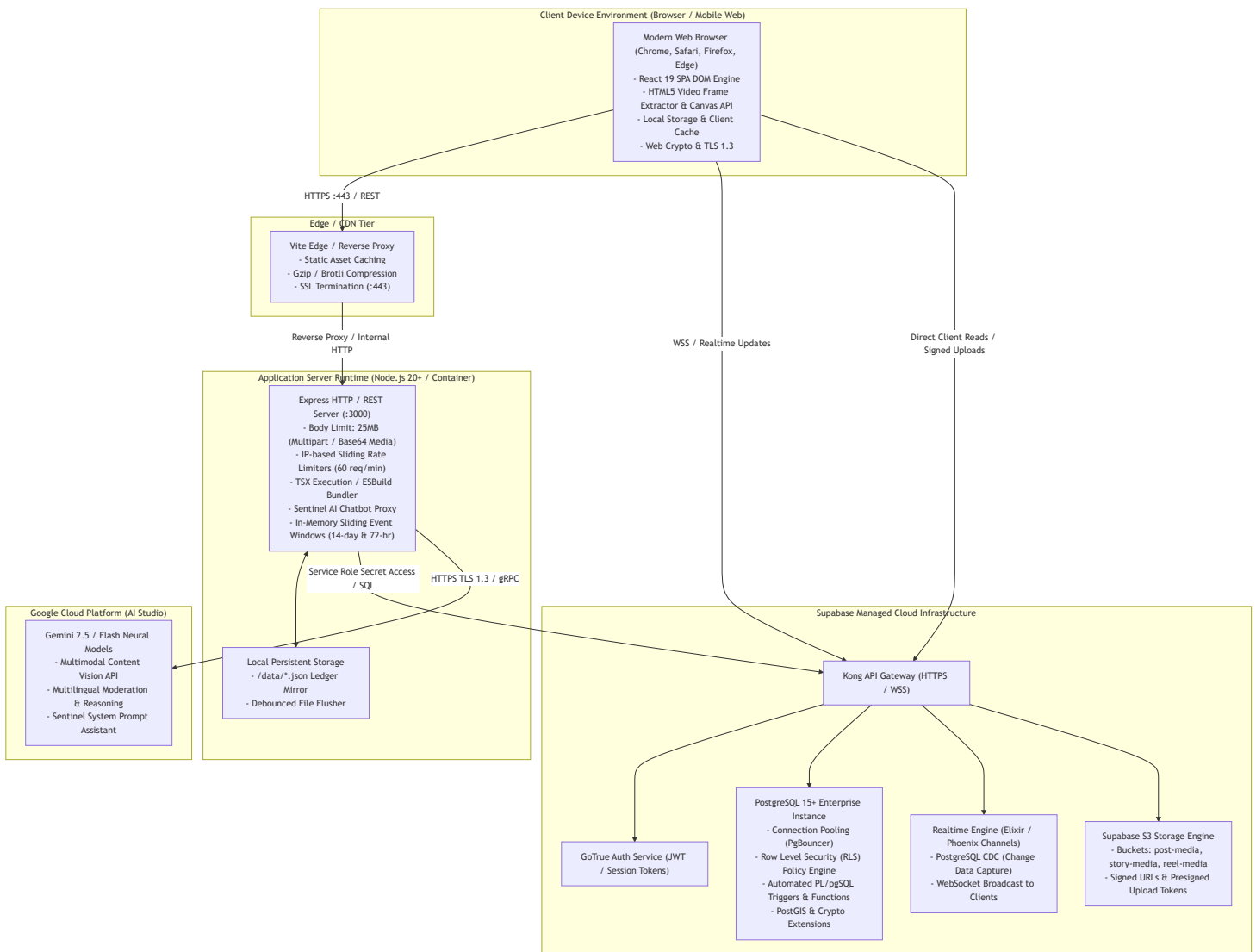
2.4 Database Entity-Relationship Diagram (ERD / Logical Data Model)

VERIXA utilizes a PostgreSQL 15 schema managed via Supabase (`supabase_schema.sql`), with 26 dedicated tables enforcing strict referential integrity, automated denormalization counter triggers, and audit logging.



2.5 Deployment & Cloud Infrastructure Diagram

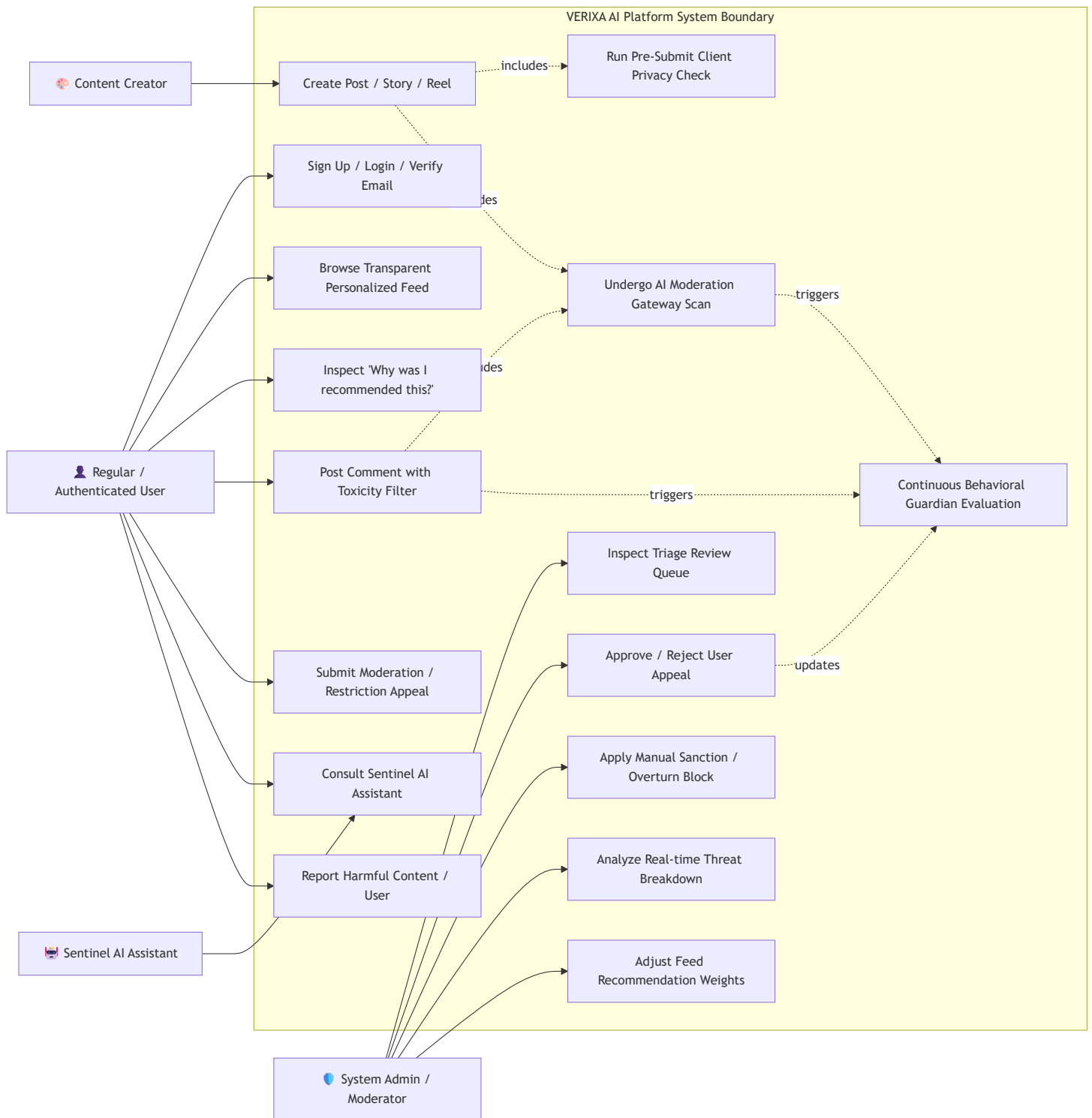
This diagram displays the execution topologies, runtime containers, hardware nodes, network boundaries, and secure protocols connecting VERIXA's distributed services.



3. Behavioral UML Diagrams

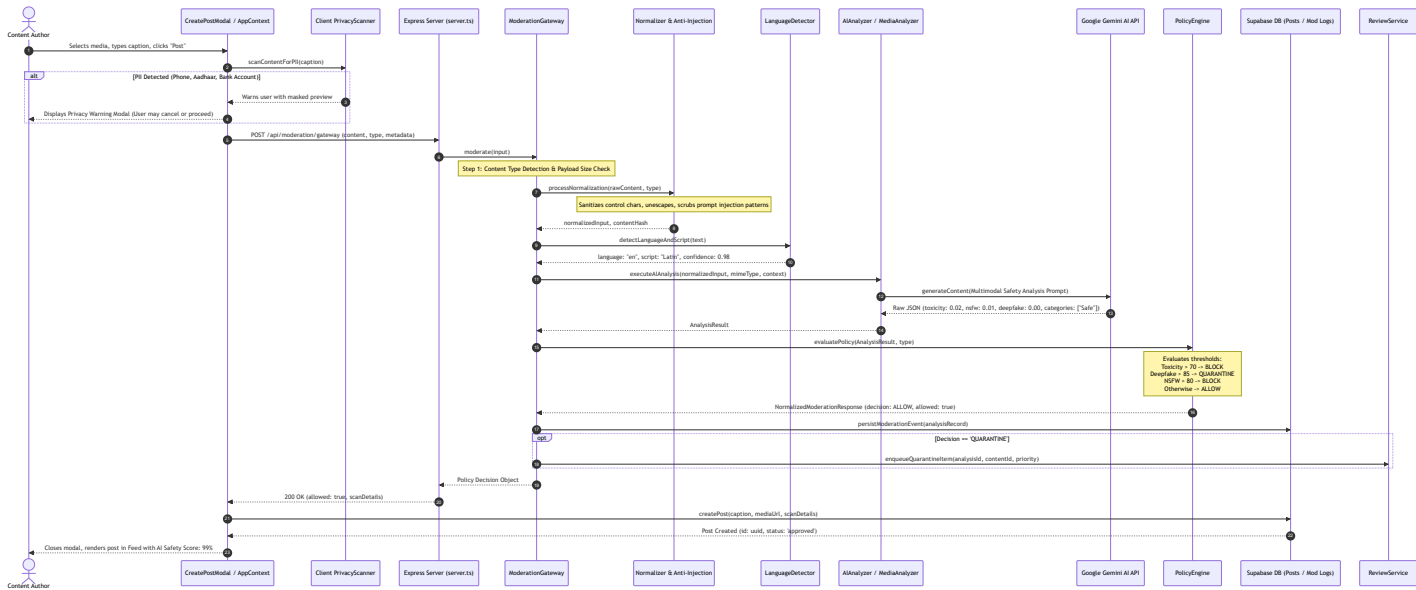
3.1 System Use Case Diagram

This diagram visualizes the behavioral interactions between system actors (**Regular User**, **Authenticated User**, **Content Creator**, **Moderator/Admin**, and **Sentinel AI Sentinel**) and the primary platform use cases.



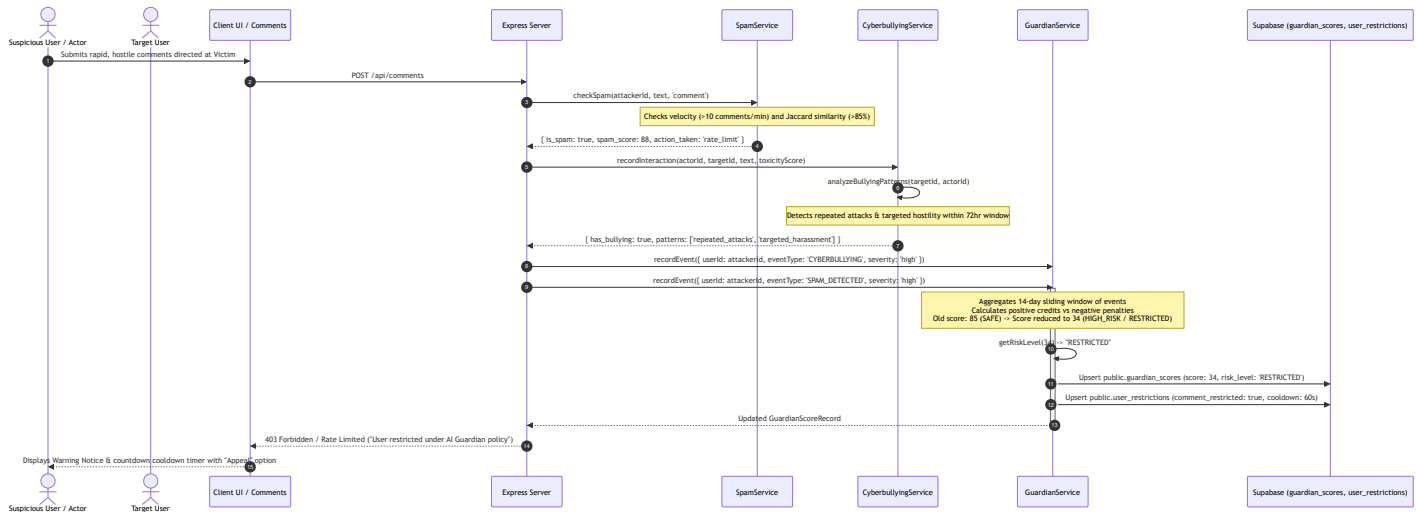
3.2 Sequence Diagram 1: Content Publishing & AI Moderation Gateway Pipeline

This diagram traces the exact synchronous execution sequence of a user publishing content through the 10-step AI Moderation Gateway.



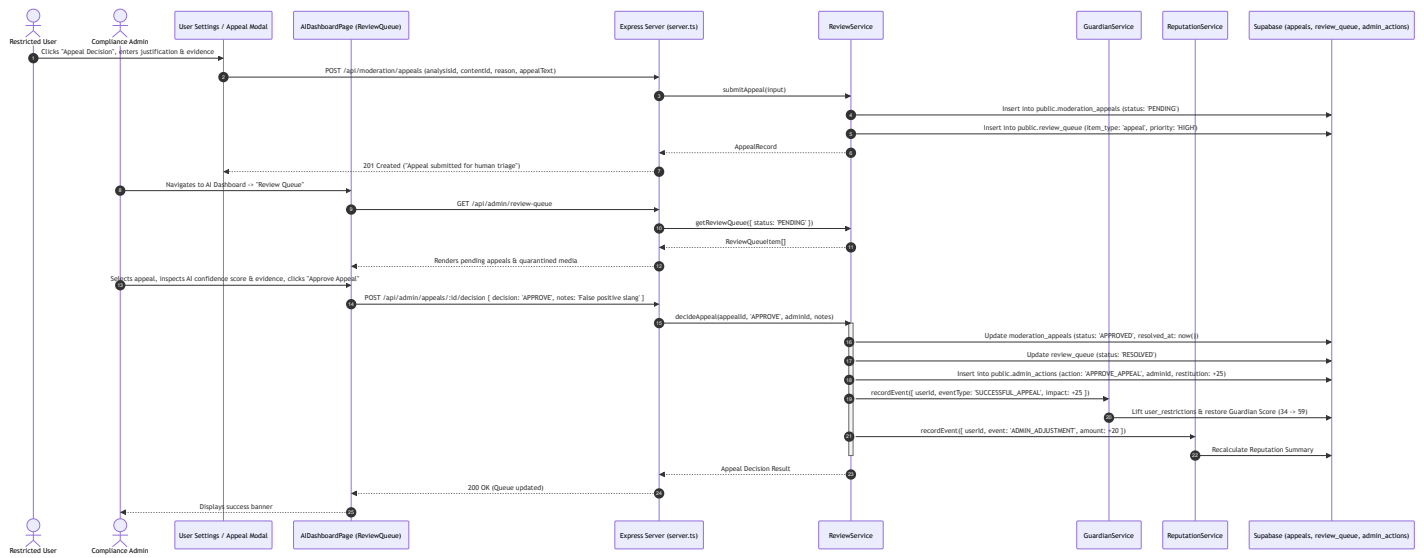
3.3 Sequence Diagram 2: Behavioral Safety & AI Guardian Risk Engine

This diagram illustrates how multi-source signals (toxicity, cyberbullying, spam, fake account heuristics) are captured and evaluated to dynamically recalculate a user's Guardian Risk Level and enforce automated rate limits or cooldowns.



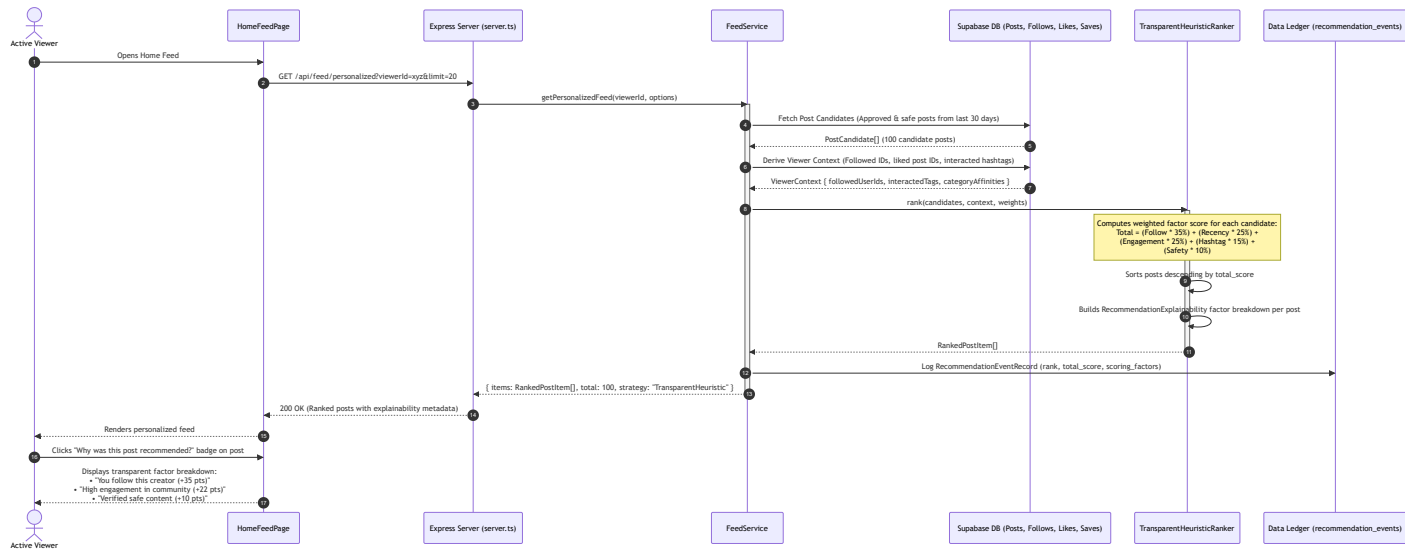
3.4 Sequence Diagram 3: Moderation Appeal & Human-in-the-Loop Review Queue

This diagram maps the lifecycle of a user appeal against a flagged post or account restriction, progressing through triage, admin adjudication, and automated score restitution.



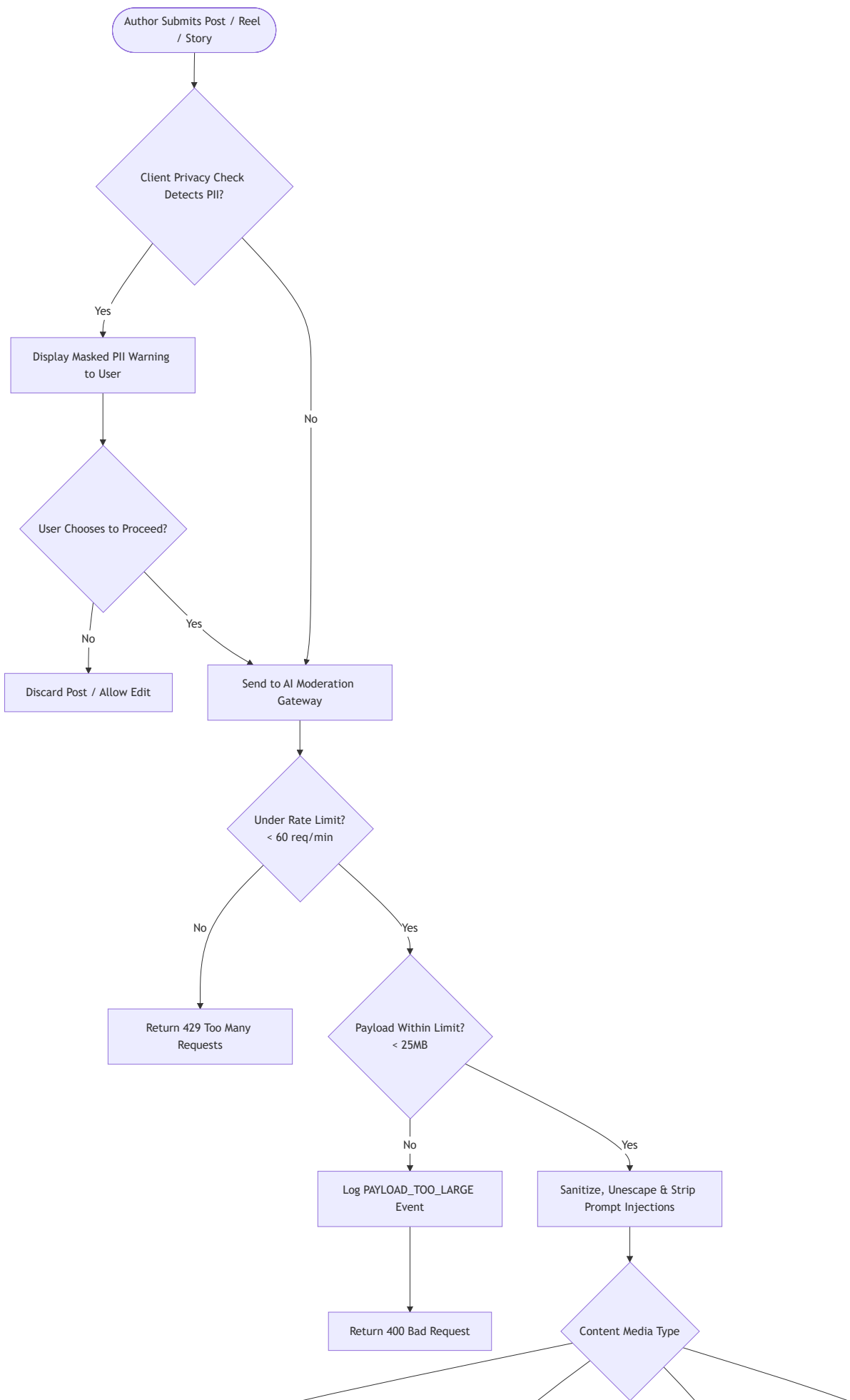
3.5 Sequence Diagram 4: Personalized Feed Recommendation & Transparent Explainability

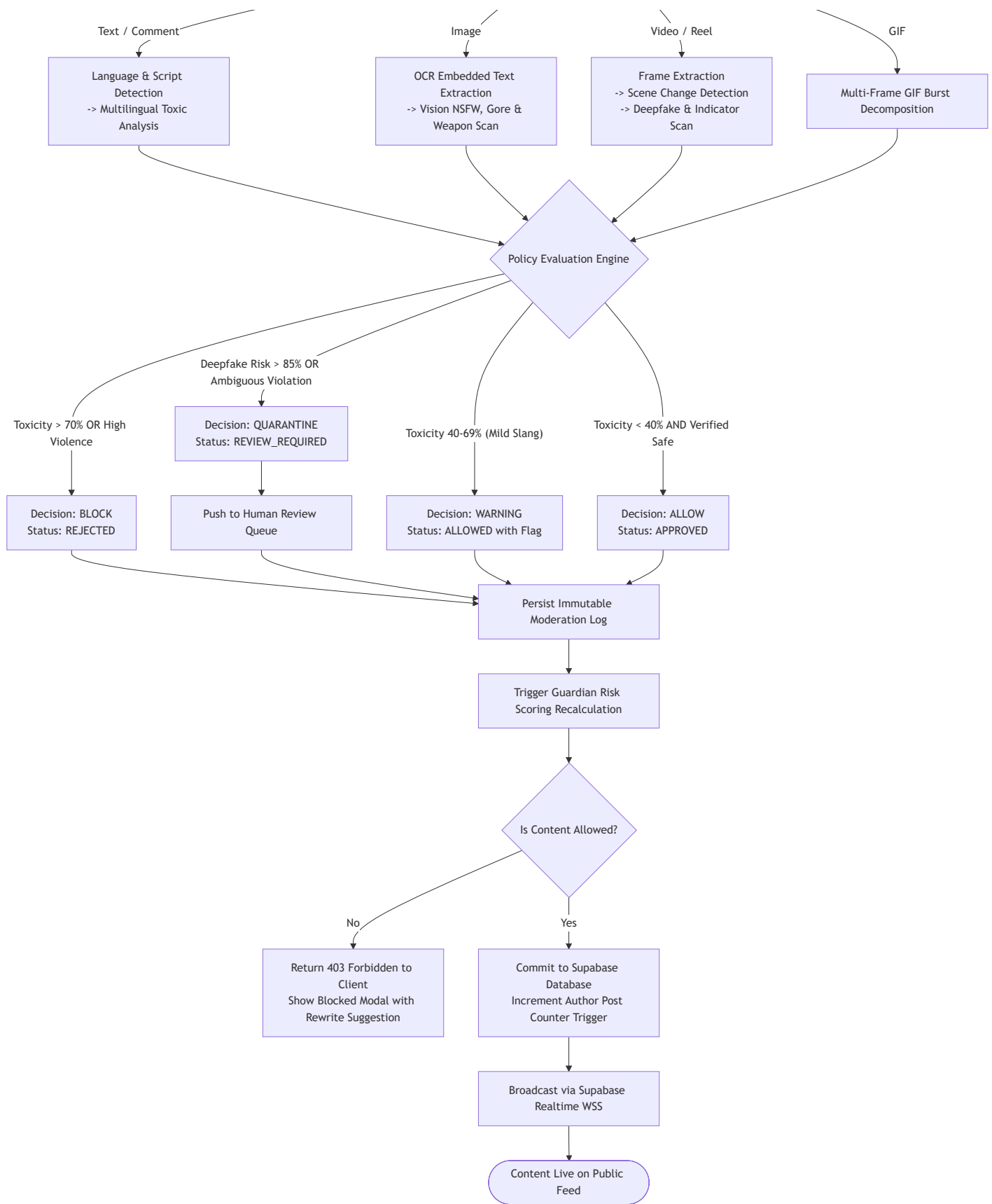
This diagram illustrates how VERIXA provides transparent algorithmic recommendation, calculating multi-factor ranking weights and generating explainable insights for every post served.



3.6 Activity Diagram: End-to-End Content Ingestion & Multi-Tier Verification Pipeline

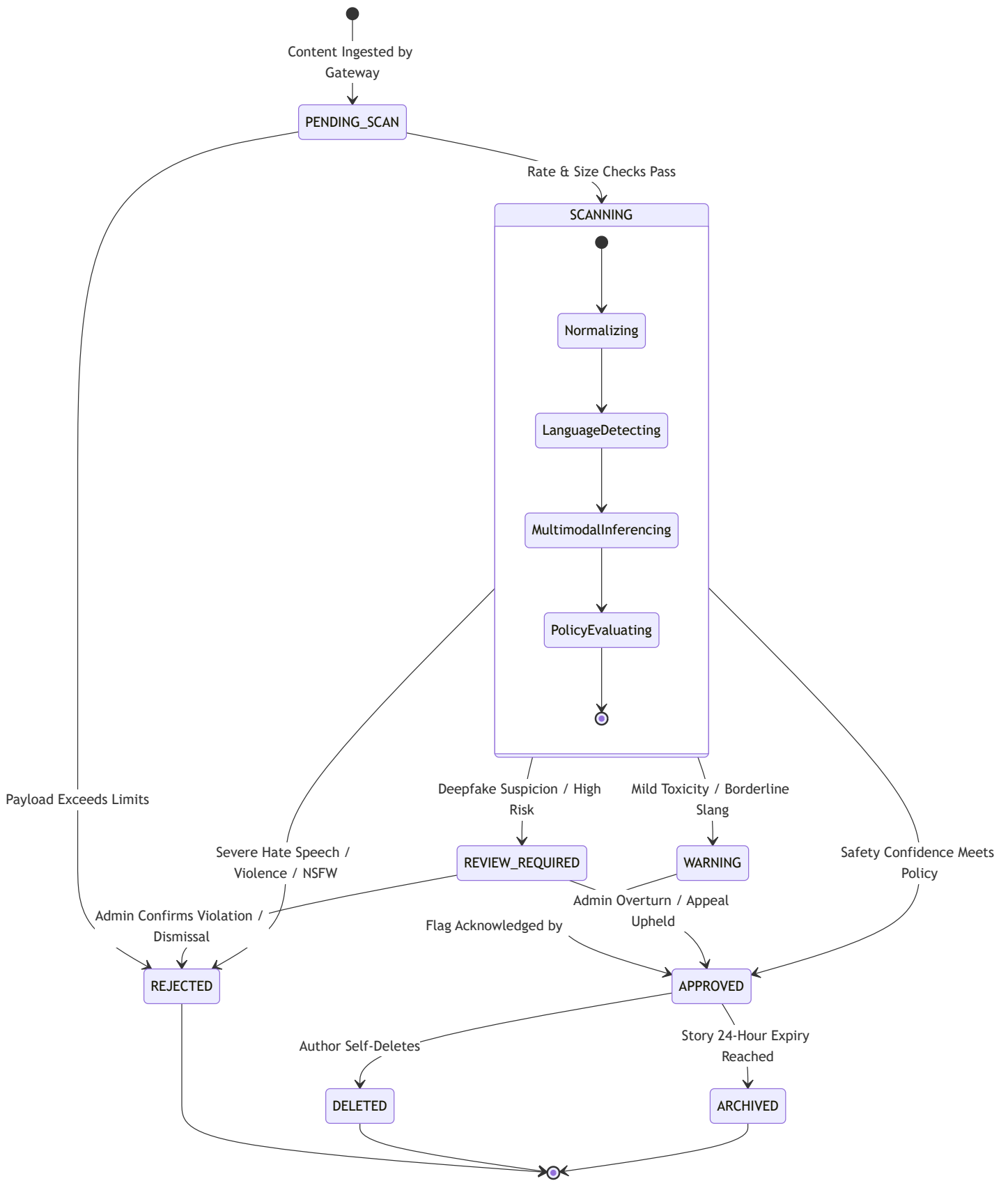
This activity flowchart details the decision nodes, branching conditions, and concurrency involved in publishing user-generated content.





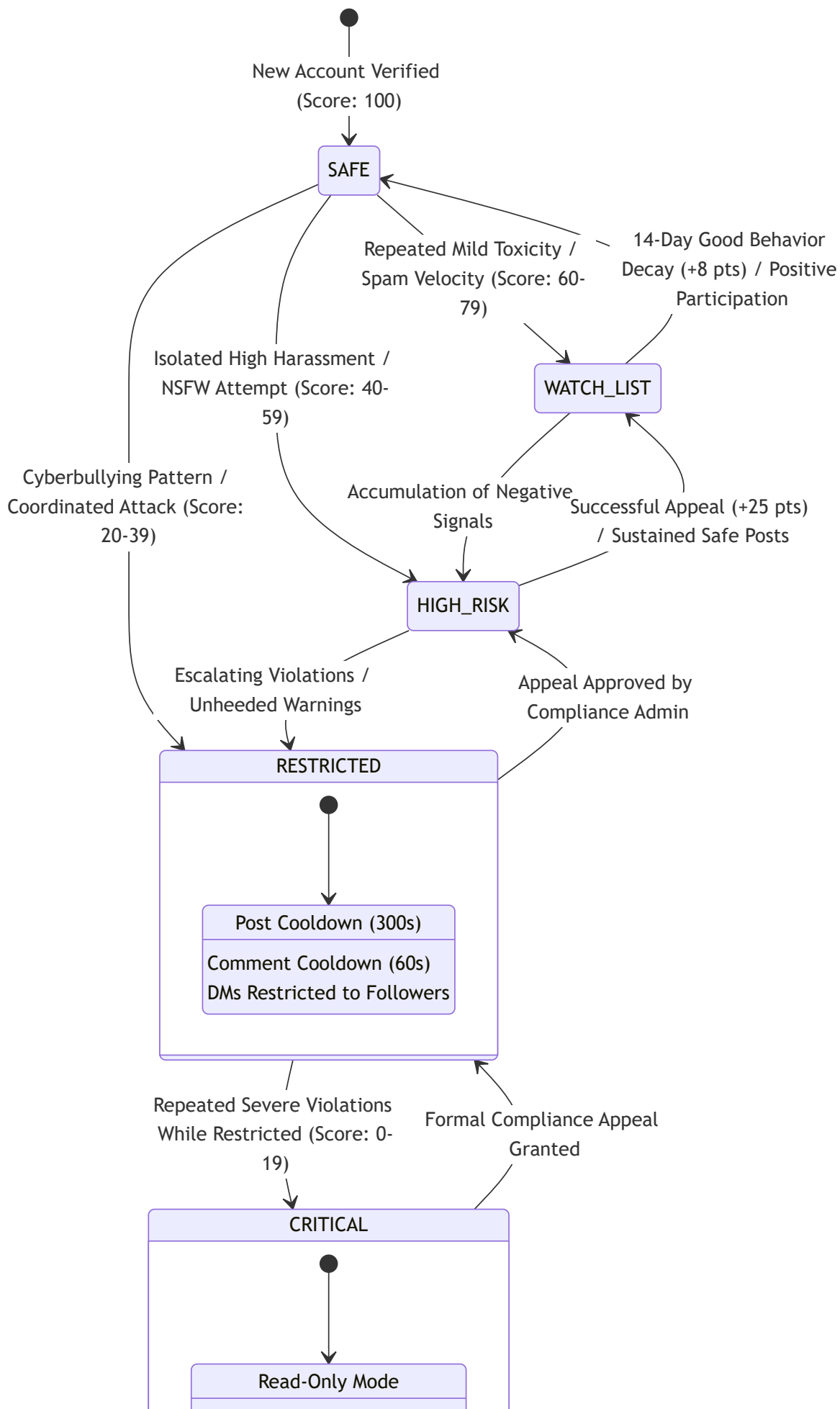
3.7 State Machine Diagram 1: Content Moderation Lifecycle

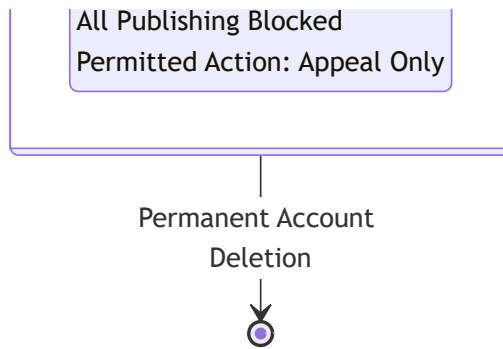
This state machine models the authoritative status transitions of any post, comment, reel, or story submitted to VERIXA.



3.8 State Machine Diagram 2: User Guardian Risk Level & Adaptive Restrictions

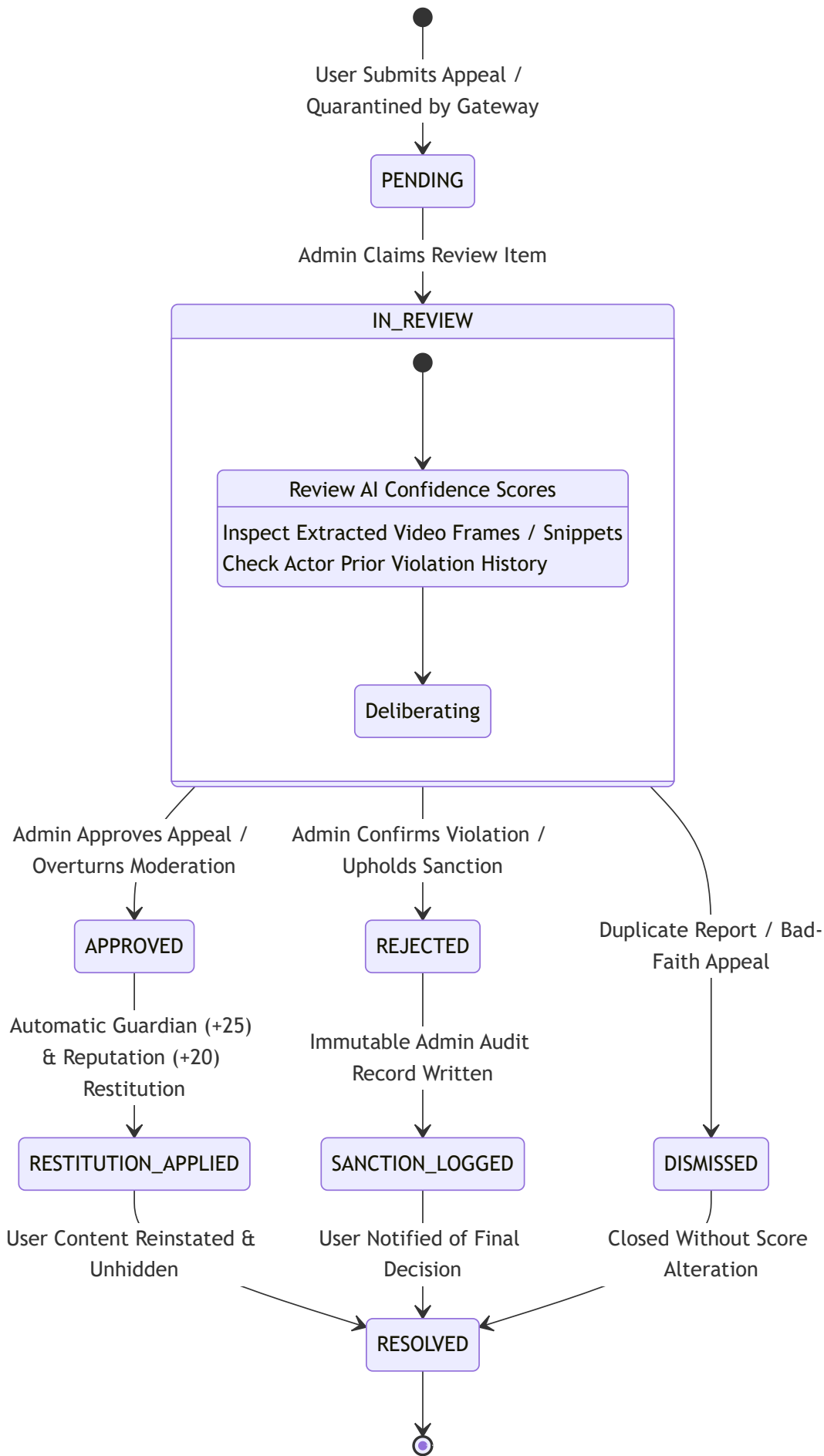
This state machine models the behavioral tier transitions of a user under the **AI Guardian Mode Engine**, detailing the actions permitted at each tier and the recovery pathways.





3.9 State Machine Diagram 3: Moderation Review & Appeal Lifecycle

This state machine traces the lifecycle of user appeals and review queue items from creation through human-in-the-loop triage and final disposition.



4. Subsystem Mapping Matrix

The following matrix provides a reference mapping across VERIXA's primary architectural subsystems, their implementation files, database tables, and corresponding UML diagrams:

Subsystem	Key Implementation Files	Supabase DB Tables	Primary UML Reference
AI Moderation Gateway	server/moderation/moderationGateway.ts server/moderation/normalizer.ts server/moderation/languageDetector.ts	public.moderation_logs	Diagram 2.1, 2.3, 3.2, 3.6, 3.7
Media Safety & Vision	server/moderation/mediaAnalyzer.ts server/moderation/videoSafetyEngine.ts server/moderation/gifExtractor.ts	public.posts , public.reels , public.stories	Diagram 2.1, 2.3, 3.6
Policy & Decision Engine	server/moderation/policyEngine.ts server/moderation/secondarySafetyRules.ts	public.moderation_logs	Diagram 2.3, 3.2, 3.6, 3.7
AI Guardian Risk Engine	server/moderation/guardianService.ts	public.guardian_scores public.guardian_events public.user_restrictions	Diagram 2.3, 2.4, 3.3, 3.8
Behavioral Intelligence	server/moderation/cyberbullyingService.ts server/moderation/spamService.ts server/moderation/fakeAccountService.ts	public.cyberbullying_events public.spam_events	Diagram 2.1, 2.3, 3.3
Privacy & PII Defense	server/moderation/privacyScanner.ts src/lib/privacyScanner.ts	Transient / Masked Logs	Diagram 2.1, 3.2, 3.6
Human Review & Appeals	server/moderation/reviewService.ts src/pages/AIDashboardPage.tsx	public.moderation_appeals public.review_queue public.admin_actions public.reports	Diagram 2.3, 2.4, 3.4, 3.9
Transparent Feed Engine	server/feed/feedService.ts server/feed/rankingStrategy.ts src/pages/HomeFeedPage.tsx	public.user_interaction_events public.served_recommendation_events	Diagram 2.1, 2.3, 2.4, 3.5
User & Social Core	src/context/AppContext.tsx src/lib/supabaseServices.ts	public.profiles , public.posts public.comments , public.likes public.follows , public.messages	Diagram 2.1, 2.2, 2.3, 2.4

Subsystem	Key Implementation Files	Supabase DB Tables	Primary UML Reference
Sentinel AI Assistant	server.ts (/api/ai-assistant) src/components/AIFloatingSentinel.tsx src/pages/SentinelAIChatbotPage.tsx	Gemini API Session / Transient	Diagram 2.1, 2.5, 3.1

End of VERIXA Unified Modeling Language (UML 2.5) System Specifications.